

PPT Guide — Stage 1: Dataset Exploration & Analysis

Built for Evaluation Round 1 (7:00-11:00 PM) — judges will be assessing progress, not a finished product

| Assumption, stated upfront

This deck covers **Stage 1 only** — since that's what you'll have finished by evaluation time. It's built so these slides can later become the *opening section* of your final full deck (Stages 1-4 + report), not a throwaway. Keep the visual style consistent with whatever you use later so it's a drop-in, not a redo.

Target: 6-7 slides, presentable in under 3 minutes. At Eval Round 1, judges are seeing many teams in a short window — dense, focused slides beat a long walkthrough.

| Slide 1: Title Slide

Content: - Project/team name - “MedhaDrishti 2026 — AI-Based Object Recognition” - Challenge: RGB-Thermal Pedestrian Detection - Team member names

Nothing more. Don't clutter the title slide — it's on screen for 5 seconds.

| Slide 2: The Problem, In Your Own Words

Don't just restate the problem statement verbatim — show you understood it.

Content (2-3 short lines max): - “Pedestrian detection fails in poor lighting/weather using RGB alone. Thermal imaging compensates — but combining both effectively, especially for small/distant pedestrians, is the open challenge.” - One line on why this matters: surveillance, autonomous systems, search & rescue

Visual: Use the sample RGB-thermal pair image from the problem statement PDF itself (Fig 1) or one of your own visualized pairs from Stage 1 output — side-by-side RGB vs thermal of the same scene is instantly clear to a judge, more effective than a paragraph of text.

| Slide 3: Dataset Overview

This is where your Stage 1 work starts appearing directly.

Content — pull straight from your `instance_statistics.csv`: | Split | Images | Pedestrian Instances | Avg per Image | | —|—|—| | Train | 1200 | [your number] | [your number] | | Val | 300 | [your number] | [your number] | | Test | 200 | [your number] | [your number] |

Keep this as an actual table on the slide, not a paragraph — judges scan tables faster than prose, and it visually signals “we did real analysis,” not just a claim.

| Slide 4: Pedestrian Scale Distribution — your most important slide in this deck

This slide does double duty: it's a required deliverable **and** it sets up your entire Stage 3 justification (why fusion/attention matters most for small objects). Make it visually strong.

Content: - A simple bar chart: Small / Medium / Large, with percentages, across train/val/test - One callout line: **“X% of pedestrians fall in the ‘small’ category (<32² pixels) — this is the core challenge our fusion strategy must address.”**

Why this matters for standing out: this slide directly connects your dataset analysis to your (upcoming) technical approach — most teams will present Stage 1 as disconnected facts; you're framing it as evidence that justifies your Stage 3 decision. Judges notice that kind of throughline.

| Slide 5: RGB-Thermal Pairs — Visual Evidence

Content: 2-3 of your best annotated visualization pairs (from `stage1_analysis/outputs/visualizations/`) — pick ones that clearly show: - A pedestrian visible in thermal but hard to see in RGB (proves *why* fusion is needed — powerful, concrete evidence) - A small/distant pedestrian correctly boxed in both modalities - If you have one: an occlusion or low-light example

Caption each image with one line explaining what it demonstrates — don't make judges guess what they're looking at. **This slide is your strongest visual moment** — real data, real boxes, real evidence, not a mockup. Spend the most polish time here.

Slide 6: Alignment Verification & Key Challenges

Content, kept brief (bullet points, not paragraphs): - ✓ RGB-Thermal alignment verified across sample pairs — [one line: "consistent" or note any misalignment found] - Key challenges identified: - Dominant small/tiny pedestrian instances - [Occlusion cases, if observed] - [Illumination variation, if observed]

This is where the "Limitations & Known Edge Cases" honesty principle from earlier applies even at this early stage — if you found real issues (misalignment, ambiguous annotations), say so plainly. It reads as rigor, not weakness.

Slide 7: What's Next (Stage 2, 3, 4 Preview)

Since this is a progress checkpoint, not a final pitch, close by showing you know where you're headed — this matters a lot at Eval Round 1 specifically, since judges are assessing trajectory, not just current state.

Content (brief, roadmap-style): - Next: benchmark RGB-only / Thermal-only / baseline QFDet (Stage 2) - Then: implement cross-modal attention fusion, targeting small-pedestrian detection improvement (Stage 3) - Finally: comparative evaluation against baseline (Stage 4)

One line closing: "Our Stage 1 analysis directly informs our fusion strategy — we're not guessing, we're building on evidence from the data itself."

Presentation delivery tips for this specific slot

- **Assign one speaker for this deck**, not a slide-by-slide handoff — with only ~3 minutes, switching speakers wastes time and breaks flow
 - **Have Slide 4 and Slide 5 memorized cold** — these are your strongest, most likely to draw follow-up questions
 - **Anticipate the obvious question:** "Why does scale distribution matter for your approach?" — have a one-sentence answer ready: small objects lose detail in RGB but often retain a heat signature in thermal, which is exactly what your fusion approach will exploit
 - **Don't apologize for only having Stage 1 done** — every team is at a similar point this early; frame it as "here's our foundation and where it's taking us," not "sorry we're behind"
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What NOT to include yet

- Don't show fusion architecture diagrams or results you don't have yet — nothing kills credibility faster than a slide that's clearly aspirational/fake at a technical hackathon
 - Don't pad with generic AI/ML stock content (definitions of object detection, generic YOLO explanations) — judges know this field, spend your limited slides on your actual work only
 - Don't include code snippets on slides — if asked, show code live on your laptop instead, slides should stay visual and high-level
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File naming/output reminder

Save this as `stage1_progress.pptx` (or your team's naming convention) in `docs/` — and reuse these exact slides as the opening section of your final deck later, just append Stages 2-4 as they complete, rather than rebuilding from scratch.